Synopsis

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Location:	4030 Kemp Ave #5 El Paso, TX 79 904-5600
Relocation:	Yes
Visa Status:	US citizen
Notice:	1 month
Attributes:	20 years semiconductor processing / characterization 12 years data science / artificial intelligence 8 years pharmaceutical development
Education:	BS Chemical Engineering 1984 MS Material Science 1999 MS Physical Chemistry 2003

Overview

> *As we approach a new era of artificial general intelligence (AGI), where artificial intelligence approaches human aptitude, I plan to leverage my skills in computer programming, optical sensors, and process control...*

With over 20 years as a Research Scientist, I have coauthored roughly 16 papers, half under the direction of John Haggerty at MIT, and the remainder with support from Klaus Bachmann at NC State. During the course of research, I have: (1) invented a patentable 'symmetric proportional control' algorithm for laser cavity optimization, stabilizing the growth of thin films (2) fabricated the first laser-induced, chemical vapor deposition, amorphous silicon solar cells, (3) developed a microwave plasma, chemical vapor deposition system, to create polycrystalline diamond from methane gas, in a regime where kinetics dominates over thermodynamics, (4) monitored the surface evolution of compound semiconductor heterostructure films, in a chemical beam epitaxy system, with plane polarized reflectance spectroscopy, pioneered by our research group.

In parallel with my academic career, I plunged into the world of macroeconomics, human behavior, and statistical analysis, through stock and futures trading. Using quantitative investment strategies, participants utilize technical analysis methods to achieve high probability trades. To handle market data, I applied digital signal processing techniques, in the spirit of John Ehlers, an Electrical Engineer from Raytheon. Along the way, I combined statistics with digital signal processing to produce highly responsive indicators, enhancing trade signal clarity. By immersing myself in the data science of financial markets, and backtesting of trading strategies, I have improved my computer programming skills, and established more techniques to deal with data interpretation.

As we approach a new era of artificial general intelligence (AGI), where artificial intelligence approaches human aptitude, I plan to leverage my skills in computer programming, optical sensors, and process control to facilitate automation within the manufacturing and office environments. The same principles which apply to measuring process parameters and regulating control valves through data acquisition can be extended to robotic autonomy. I invite you to visit my personal website to view current literature and computational projects. If you have any concerns, feel free to contact me.

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Profile

Chemical Engineer seeking a Process Engineer, Research & Development, or Data Science role, applying my core skills in:

chemical vapor deposition	plasma chemistry	heteroepitaxial structures
molecular beam epitaxy	laser excitation	synthetic biology
reactive ion etching	optical characterization	python language
photolithography	electrochemical methods	statistical analysis
metallization techniques	additive manufacturing	process control

Thesis

Real Time Reflectometry of Ga–based Compound Semiconductor Films on Silicon during Plasma Enhanced Molecular Beam Epitaxy, NCSU Materials Science Dept: **1999**.

Clifton Strengths

Character

Strategic	faced with any given scenario, can quickly spot the relevant patterns.
Learner	have a great desire to learn and want to continuously improve.
Ideation	able to find connections between seemingly disparate phenomena.
Futuristic	inspired by the future and what could be.
Self-Assurance	possess an inner compass to instill confidence in decision making.

Experience

Prompt Engineer, Outlier AI: San Francisco, CA (10/24 to present)

- Generate prompt questions to evaluate AI model responses to python programming tasks.
- Write and rewrite AI model responses to prompt questions with accurate python code.

Network Member, Gerson Lehrman Group: Austin, TX (4/20 to present)

- Appear as an expert witness in a patent lawsuit regarding diamond thin films.
- Provide scientific insight in a myriad of semiconductor issues.

Quantum Trader, Independent (5/84 to present)

- Broaden the scope to include both fundamental evaluation and technical analysis.
- Combine statistics with digital signal processing to produce indicators with better trade signal clarity.
- Implement python programming to learn hedge fund strategies.
- Evaluate fundamental aspects of technology and pharmaceutical sectors.
- Deploy venture capital principles to choose lucrative issues, including initial pubic offerings.

Laboratory Technician, Genesis Biotechnology Group: Hamilton, NJ (7/20 to 10/21)

- Process COVID-19 nasal swabs in Biological Safety Level 2 hoods under CDC guidelines.
- Extract nucleic acid samples from blood, urine, spinal, and other body fluids using Vacuum Filtration or Magnetic Bead separation.
- Perform static Polymerase Chain Reaction (PCR) and dynamic PCR (qPCR).
- Provide in-house quality control, identifying errors in: sample placement, robot operation, and SOPs.
- Apply advanced process engineering techniques to make Biomek robot extractions more reliable.
- Modify machine code to customize Biomek robot extractions.

Research Assistant, Maine Chemistry Dept: Orono, ME (8/03 to 5/06)

- Synthesize aurophilic gold compounds, a potential drug candidate for arthritis.
- Apply cyclic voltammetry to investigate catalytic activity in gold compounds.
- Induce electrochemiluminescence in a ruthenium compound for DNA analysis.

Teaching Assistant, Rutgers Chemistry Dept: New Brunswick, NJ (1/00 to 1/03)

- Present lab techniques to students on all undergraduate levels, including PChem.
- Pass the cumulative exam, the written portion of a PhD degree.

Research Assistant, NCSU Materials Science Dept: Raleigh, NC (1/87 to 12/99)

- Develop a microwave plasma, chemical vapor deposition system, to create polycrystalline diamond from methane gas, in a regime where kinetics dominates over thermodynamics.
- Achieve unique ellipsoidal plasma, energizing the surface, relative to spherical geometry.
- Design a radio frequency nitrogen plasma source for GaN film growth.
- Monitor the surface evolution of compound semiconductor heterostructure films, in a chemical beam epitaxy system, with plane polarized reflectance spectroscopy.
- Derive substrate temperature from plane polarized reflectance intensity.

Research Specialist, MIT Advanced Energy Materials Lab: Cambridge, MA (11/84 to 1/87)

- Invent a new approach for process control to optimize laser power.
- Write a Pascal based data acquisition program for DOS environment in 1986, long before LabView enters the Windows market.
- Analyze optical signals from a ceramic powder reaction chamber, leading to a computer monitoring scheme, which replaces a human operator.
- Construct interferometer to measure film thickness, providing a realtime signal, to calibrate growthrate.
- Refine process control loop to stabilize laser power, producing a steady deposition rate with reliable material properties.
- Collect in-situ stress measurements of growing films, through deflection of an optical laser, as sample curvature evolves.
- Grow the first laser-induced, chemical vapor deposition, amorphous silicon solar cell.

Education

MS *Physical Chemistry*	Rutgers: New Brunswick, NJ	Jan 2003
MS *Material Science*	North Carolina State: Raleigh, NC	unofficial
BS *Chemical Engineering*	Texas A&M: College Station, TX	May 1984
HS *Diploma*	Waltham High: Waltham, MA	Jun 1979

Honor

Bausch & Lomb Science Award